The numerical value of the length of the vector is irrelevant; we can just pick it to be a convenient number, knowing that it will remain constant. Let's pick it to be one: every wave function is a vector of length one. The vector itself is just like the hypotenuse of a right triangle, with the components forming the shorter sides. So from Pythagoras's theorem, we have a simple relationship: the squares of the amplitudes add up to unity, $|a|^2 + |b|^2 = 1$.

That's the simple geometric fact underlying the Born rule for quantum probabilities. Amplitudes themselves don't add up to one, but their squares do. That is exactly like an important feature of probability: the sum of probabilities for different outcomes needs to equal one. (Something has to happen, and the total probability of all exclusive somethings adds up to unity.) Another rule is that probabilities need to be non-negative numbers. Once again, amplitudes squared fit the bill: amplitudes can be negative (or complex), but their squares are non-negative real numbers.

So even before thinking too hard, we can tell that "amplitudes squared" have the right properties to be the probabilities of outcomes—they are a set of non-negative numbers that always add up to one, because that's the length of the wave function. This is at the heart of the whole matter: the Born rule is essentially Pythagoras's theorem, applied to the amplitudes of different branches. That's why it's the amplitudes squared, not the amplitudes themselves or the square root of the amplitudes or anything crazy like that.

The vector picture also explains the uncertainty principle in an elegant way. Remember that spin-up electrons split fifty-fifty into right-and left-spinning electrons when they passed through a subsequent horizontal magnet. That suggests that an electron in a spin-up state is equivalent to a superposition of spin-right and spin-left electron states, and likewise for spin-down.